ASSIL HAMDI MEMORIAL MATHEMATICS OLYMPIAD - 2026

Official Examination Paper - 4 Hours Strict Duration

Rules: 1. Legible handwriting required. 2. Clear geometric schemas mandatory.
3. Zero tolerance for AI or plagiarism: Result will be 0 points and account ban.

Problem 1 (Geometry - 7 Points):

Let ABC be an acute triangle with circumcircle Omega. Points D and E lie on sides AB and AC respectively. Prove that the radical axis of the two circles passes through...

Problem 2 (Number Theory - 7 Points):

Find all pairs of positive integers (x, y) such that 2^x + 3^y is a perfect square.

Problem 3 (Functional Equations - 7 Points):

Find all functions f: R -> R satisfying f(xf(y) + f(x)) = 2f(x) + xy for all reals x, y.

Problem 4 (Combinatorics - 7 Points):

A 2026 x 2026 grid is tiled by dominoes. Prove that the number of symmetric tilings is...